UML Component Diagram

Python Calculator Module - Simple Layered Architecture

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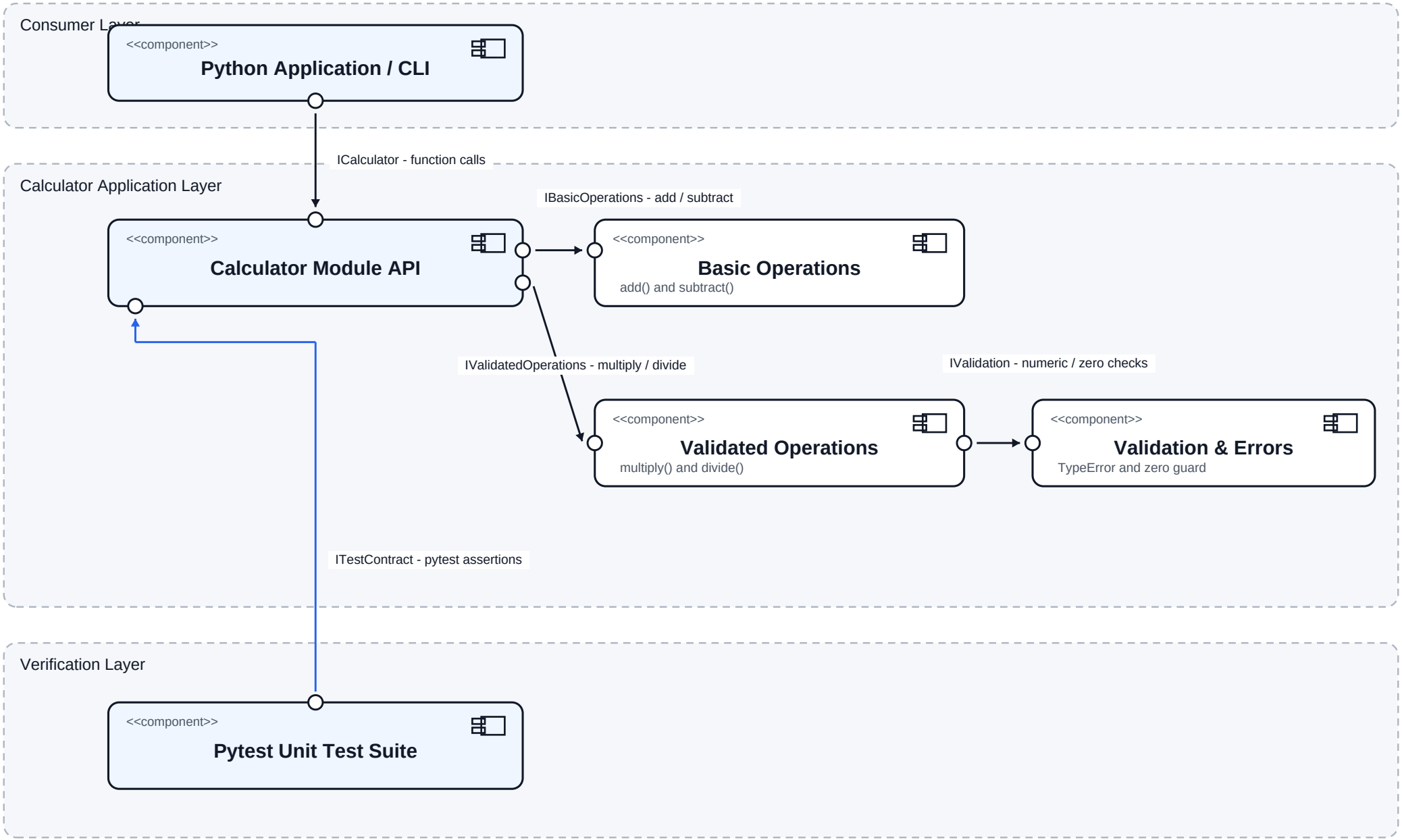

Legend ○ provided interface) required interface → request or dependency flow

6 components | 5 labeled interfaces

Architecture Justification

Python Calculator Module

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Architecture Choice

The calculator uses a simple layered component architecture. The consumer layer invokes a stable public API, the application layer performs arithmetic and validation, and the verification layer checks the same public behavior through pytest.

Why It Fits

Each component has one clear responsibility. Client code is isolated from implementation details, basic operations stay small, and validated operations can evolve without changing how callers use the module. This is appropriately lightweight for a compact Python project.

Reliability Advantage

Multiplication and division reject non-numeric inputs before performing arithmetic. Division also prevents a zero denominator and raises an explicit error. Keeping these checks at the operation boundary produces predictable failures and protects callers from unclear runtime behavior.

Maintainability and Testing Benefit

The public functions provide a small testable contract. Pytest can verify results and expected exceptions independently, while future operations such as power and square root can be added as new components without disturbing existing callers. The separation also makes defects easier to locate.

Performance Benefit

The design uses direct in-process function calls and constant-time arithmetic, so it introduces no network, database, or serialization overhead. Validation is limited to inexpensive type and boundary checks, keeping the module responsive while still improving correctness.

Implementation Mapping

The Calculator Module API, arithmetic operations, and validation behavior are implemented in **src/calculator.py**. The verification component is represented by **tests/unit/test_calculator.py** and the project pytest configuration.